

Appendix:

A. Rank of 1D Hankel is determined by support of 1D signal in image domain

MRI acquisition is usually modelled as sampling at the Fourier transform of image function $\rho(x)$ at Nyquist rate (Haldar and Setsompop, 2020):

$$\tilde{\rho}[n] = \int_{-B}^B \rho(x) e^{-i2\pi nx/2B} dx, \quad n = 0, 1, \dots, 2N-2 \quad \text{Eq. (A1)}$$

where $\tilde{\rho}[n]$ is the n^{th} sample in the Fourier domain (k-space), $\rho(x)$ is a continuous, 1D, complex-valued image function with $x \in \mathbb{R}$, and the interval $[-B, B] \subset \mathbb{R}$ is known as the field of view. $2N-1$ samples are acquired in total.

We rewrite Eq. (A1) into a discrete form:

$$\tilde{\rho}[n] = \sum_{r=0}^R \rho(x_r) z_r^n, \quad 0, 1, \dots, 2N-2, \quad \text{Eq. (A2)}$$

where $\rho(x_r), r=1, 2, \dots, R$ are the non-zero values in $\rho(x)$, $z_r = e^{-i2\pi x_r/2B}$. We can see that the k-space signal $\tilde{\rho}[n]$ is a superposition of complex exponential signals. We lift a Hankel matrix $\mathbf{H} \in \mathbb{C}^{N \times N}$ from $\tilde{\rho}[n]$ as:

$$\mathbf{H}_{ij} = \tilde{\rho}[i+j], \quad i, j = 0, 1, \dots, N-1. \quad \text{Eq. (A3)}$$

Perform a Vandermonde decomposition on $\mathbf{H}$, and we get

$$\mathbf{H} = \mathbf{E} \mathbf{\Sigma} \mathbf{E}^T = \begin{bmatrix} 1 & \dots & 1 \\ z_1 & \dots & z_1 \\ \vdots & \dots & \vdots \\ z_1^{N-1} & \dots & z_1^{N-1} \end{bmatrix} \begin{bmatrix} \rho(x_1) & & \\ & \ddots & \\ & & \rho(x_R) \end{bmatrix} \begin{bmatrix} 1 & z_1 \dots & z_1^{N-1} \\ \vdots & \vdots & \vdots \\ 1 & z_1 \dots & z_1^{N-1} \end{bmatrix}. \quad \text{Eq. (A4)}$$

The rank of $\mathbf{H}$ is R , the number of non-zeros in image $\rho(x)$. Thus, $\mathbf{H}$ is rank-deficient, when image $\rho(x)$ has a good limited support ($R < N$).

We use a top example (Fig. A1) to verify the above derivations. In an ideal noise-free situation, we construct 1D Hankel matrices from the 1st, 64th, 128th k-space rows, $\tilde{\rho}_1[n]$, $\tilde{\rho}_{64}[n]$, $\tilde{\rho}_{128}[n]$, respectively. Since they have approximately equal image support (Fig. A1(b)), the ranks of constructed Hankel matrices are all equal to 29 (Fig. A1(d)). When $\Delta N / N$ decrease to 0.3 (by reducing the width of the brain), the ranks of all constructed Hankel matrices are all decreased to about 17.

However, in the unnormalized singular values distributions (Figs. A1(e) and (g)), we can see the significant strength difference between Hankel matrices from the 1st, 64th, 128th k-space rows, indicating their different resistance to noise.

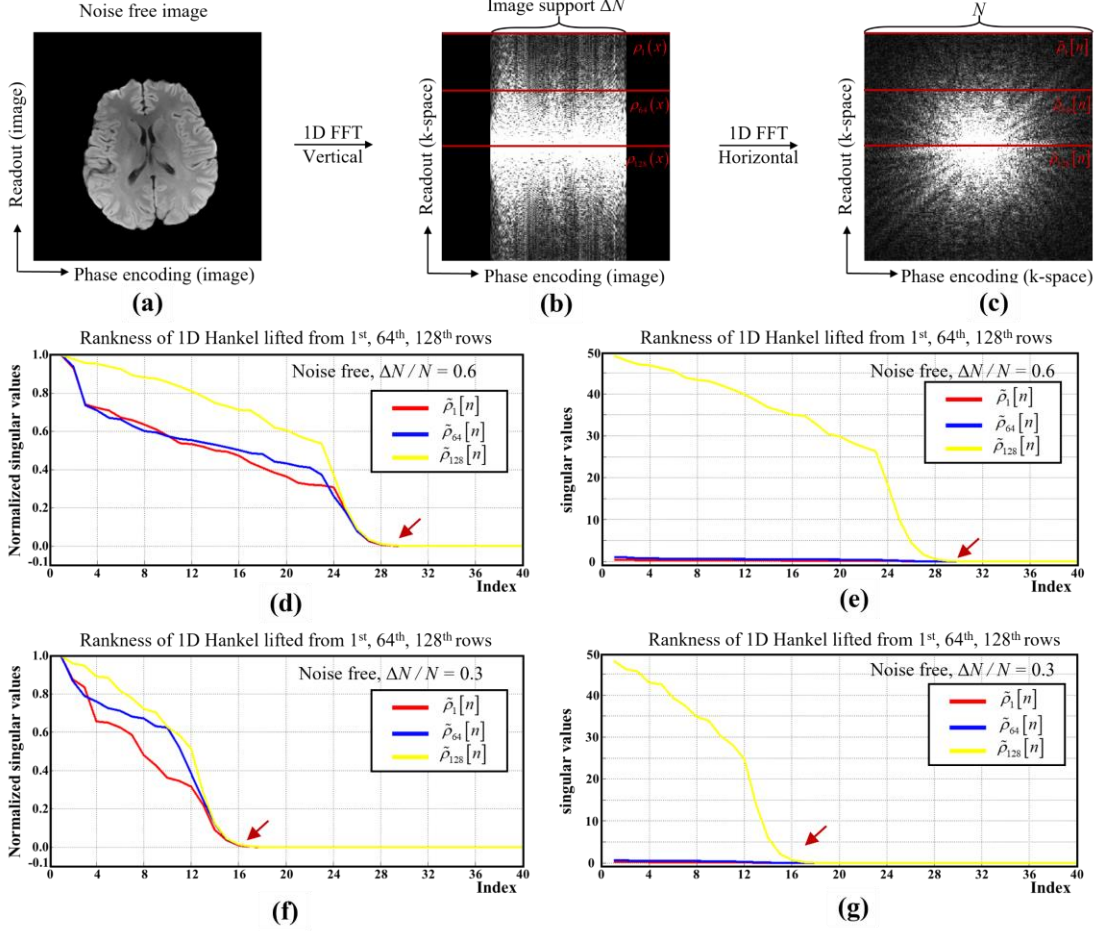


Fig. A1. Illustration of low-rankness of 1D Hankel in noise-free situation. (a) is a noise free brain image. (b) is the 1D FFT (vertical) result of (a), and the image support of each row in phase encoding direction is about ΔN . (c) is the k-space of (a), and $\tilde{\rho}_1[n]$, $\tilde{\rho}_{64}[n]$, $\tilde{\rho}_{128}[n]$ are 1st, 64th, 128th k-space row, respectively. (d) and (e) are the normalized and unnormalized singular values distributions of Hankel matrices lifted from $\tilde{\rho}_1[n]$, $\tilde{\rho}_{64}[n]$, $\tilde{\rho}_{128}[n]$, respectively ($\Delta N / N = 0.6$). (f) and (g) are the normalized and original singular values distributions of Hankel matrices lifted from $\tilde{\rho}_1[n]$, $\tilde{\rho}_{64}[n]$, $\tilde{\rho}_{128}[n]$, respectively ($\Delta N / N = 0.3$).

B: Rank of 1D Hankel is significantly affected by noise

When the noise is added to k-space to 30 dB (Figs. A2 (a) and (c)), the Hankel matrices from $\tilde{\rho}_1[n]$, $\tilde{\rho}_{64}[n]$ become full-rank, while Hankel from $\tilde{\rho}_{128}[n]$ is rank-deficient. When the noise is added to 15 dB (Figs. A2 (b) and (d)), Hankel from $\tilde{\rho}_{128}[n]$ is still rank-deficient, showing good resistance to noise.

Thus, DONA encounters the performance drop due to the same low-rank constraint applied to Hankel matrices with different 1D low-rankness.

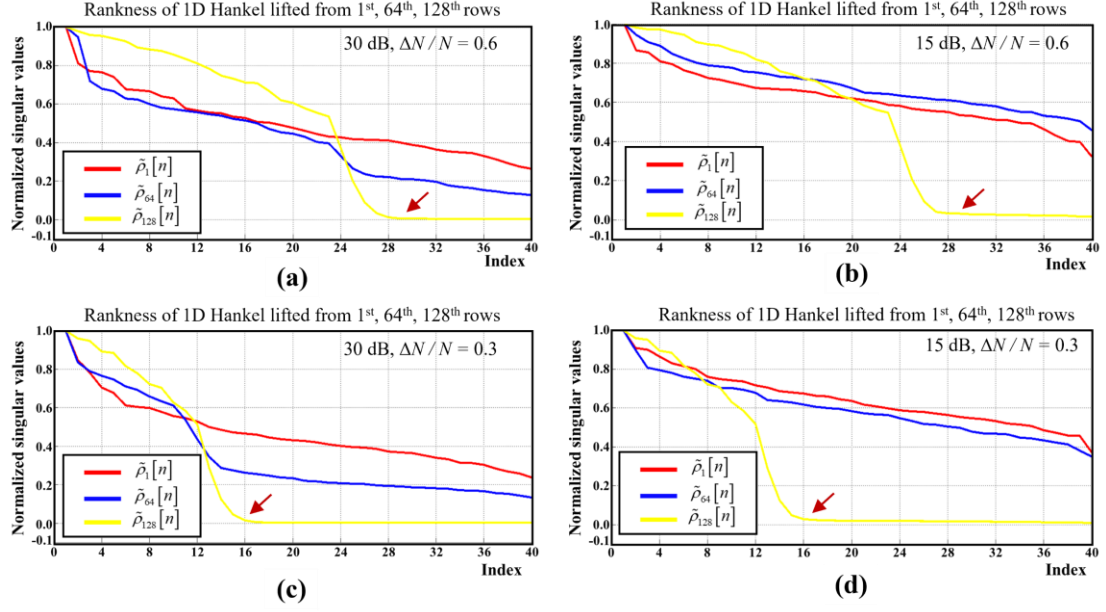


Fig. A2. Illustration of low-rankness of 1D Hankel with different noise levels. The noise is added in the k-space. (a)-(d) are the normalized singular values distributions of Hankel matrices lifted from $\tilde{\rho}_1[n]$, $\tilde{\rho}_{64}[n]$, $\tilde{\rho}_{128}[n]$, respectively.

C: The detail description on the process of pre-reconstruction

We get the full center k-space data of each shot (rather than channel) from the under-sampled multi-shot EPI data by pre-reconstruction. Detailed process is described as:

- 1) The acquired multi-shot DWI datasets (including $\mathbf{B}_0$ and DWI data) is preprocessed to account for EPI odd-even trajectory shifts arising from eddy currents.
- 2) Crop the $L \times L$ center k-space $\mathbf{Y}^{center} \in \mathbb{C}^{L \times L \times H \times J}$ from the sampled DWI data $\mathbf{Y} \in \mathbb{C}^{N \times M \times H \times J}$. N , M , H , and J are number of phase encoding, readout, channels, and shots, respectively. Note that, both $\mathbf{Y}$ and $\mathbf{Y}^{center}$ are under-sampled now.
- 3) The coil sensitivity maps $\mathbf{C} \in \mathbb{C}^{N \times M \times H}$ is estimated from $\mathbf{B}_0$. Resize the $\mathbf{C}$ into $\mathbf{C}^{center} \in \mathbb{C}^{L \times L \times H}$.
- 4) With $\mathbf{C}^{center}$, $\mathbf{Y}^{center}$, and corresponding sampling mask, we can use any multi-shot DWI reconstruction method to obtain the fully-sampled k-space of each shot image $\mathbf{X}^{center} \in \mathbb{C}^{L \times L \times J}$. The $\mathbf{X}^{center}$ can provide relatively reliable strong subspace for DONATE. In our work, PAIR is used for this pre-reconstruction. Note that, this pre-reconstruction is very fast, since L is very small, compared to N and M . We obtain only the k-space of each shot image, rather than each channel image.

D: The mathematical derivations of 2D block Hankel matrix in PAIR and 1D low-rank Hankel matrix in DONA and DONATE:

1) Mathematical derivations of 2D block Hankel matrix in PAIR

Substitute the j^{th} shot image $\mathbf{I}_j = \mathbf{P}_j \mathbf{m}$ and the complex conjugate of the shot image

$\mathbf{I}_j^* = (\mathbf{P}_j \mathbf{m})^* = \mathbf{P}_j^* \mathbf{m}^* = \mathbf{P}_j^* \mathbf{m}$ into $(\mathbf{P}_j \mathbf{P}_j^*) \mathbf{m} = (\mathbf{P}_j^* \mathbf{P}_j) \mathbf{m}$, we get:

$$\mathbf{P}_j \mathbf{I}_j^* = \mathbf{P}_j^* \mathbf{I}_j. \quad \text{Eq. (D1)}$$

Transform the left and right of Eq. (C1) into k-space:

$$\tilde{\mathbf{P}}_j \otimes \mathbf{X}_j^* = \tilde{\mathbf{P}}_j^* \otimes \mathbf{X}_j, \quad \text{Eq. (D2)}$$

where $\otimes$ is convolution, $\mathbf{X}_j \in \mathbb{C}^{N \times M}$ and $\tilde{\mathbf{P}}_j \in \mathbb{C}^{N \times M}$ are the Fourier transform of $\mathbf{I}_j$ and $\mathbf{P}_j$. Rewrite Eq. (C.2) into a multiplication form:

$$\sum_{(p,q) \in \mathbb{Z}} \tilde{\mathbf{P}}_j(p,q) \mathbf{X}_j^*(-x-p, -y-q) - \sum_{(p,q) \in \mathbb{Z}} \tilde{\mathbf{P}}_j^*(p,q) \mathbf{X}_j(x-p, y-q) = 0, \quad \text{Eq. (D3)}$$

where (x, y) is the coordinate of $\mathbf{X}_j$, $x \in [0, N]$ and $y \in [0, M]$. $(p, q) \in \mathbb{Z}$ is the coordinate of $\tilde{\mathbf{P}}_j$ and $\mathbb{Z}$ is the region where $\tilde{\mathbf{P}}_j(p, q)$ is non-zero.

The nonzero value of $\tilde{\mathbf{P}}_j$ in k-space is concentrated in the limited support due to the smoothness of $\mathbf{P}_j$ in the image. Thus, $\mathbb{Z}$ can be approached by a radius R circle region $\mathbb{Z}_R$. With this approximate representation, Eq. (3) holds an annihilation relationship (Haldar, 2013):

$$\sum_{(p,q) \in \mathbb{Z}_R} \mathbf{X}_j^*(-x-p, -y-q) \tilde{\mathbf{P}}_j(p, q) - \sum_{(p,q) \in \mathbb{Z}_R} \mathbf{X}_j(x-p, y-q) \tilde{\mathbf{P}}_j^*(p, q) \approx 0, \quad \text{Eq. (D4)}$$

Split the real and imaginary parts of Eq. (D4):

$$\begin{aligned} & \sum_{(p,q) \in \mathbb{Z}_R} (\mathbf{X}_j^r(-x-p, -y-q) - \mathbf{X}_j^i(-x-p, -y-q)) (\tilde{\mathbf{P}}_j^r(p, q) + \tilde{\mathbf{P}}_j^i(p, q)) \\ & - \sum_{(p,q) \in \mathbb{Z}_R} (\mathbf{X}_j^r(x-p, y-q) + \mathbf{X}_j^i(x-p, y-q)) (\tilde{\mathbf{P}}_j^r(p, q) - \tilde{\mathbf{P}}_j^i(p, q)) \approx 0, \end{aligned} \quad \text{Eq. (D5)}$$

where the superscript r and i represent real and imaginary part, respectively. The real and imaginary of Eq. (D5) could be split into two equations, and the matrix multiplication is:

$$\begin{cases} (\mathbf{X}_j^{r+} - \mathbf{X}_j^{r-}) \hat{\mathbf{P}}_j^r + (\mathbf{X}_j^{i+} - \mathbf{X}_j^{i-}) \hat{\mathbf{P}}_j^i = \mathbf{0}, \\ (\mathbf{X}_j^{i+} + \mathbf{X}_j^{i-}) \hat{\mathbf{P}}_j^r + (-\mathbf{X}_j^{r+} - \mathbf{X}_j^{r-}) \hat{\mathbf{P}}_j^i = \mathbf{0}, \end{cases} \quad \text{Eq. (D6)}$$

$$\begin{cases} \mathbf{X}_j^{r+}(e, f) = \mathbf{X}_j^r(x_e - p_f, y_e - q_f), \\ \mathbf{X}_j^{r-}(e, f) = \mathbf{X}_j^r(-x_e - p_f, -y_e - q_f), \\ \mathbf{X}_j^{i+}(e, f) = \mathbf{X}_j^i(x_e - p_f, y_e - q_f), \\ \mathbf{X}_j^{i-}(e, f) = \mathbf{X}_j^i(-x_e - p_f, -y_e - q_f), \\ \hat{\mathbf{P}}_j^r(f, 1) = \tilde{\mathbf{P}}_j^r(p_f, q_f), \quad \hat{\mathbf{P}}_j^i(f, 1) = \tilde{\mathbf{P}}_j^i(p_f, q_f), \end{cases} \quad \text{Eq. (D7)}$$

where $e \in [1, (N-R) * (M-R)]$ and $f \in [1, N_R]$, N_R is the number of elements in $\mathbb{Z}_R$

(Huang et al., 2020). Finally, we get $\mathcal{P}$ the operator that converts the multi-shot k-space data into a Hankel matrix (in the under-braces). The Eq. (D8) implies that this Hankel matrix is rank-deficient (Haldar, 2013) and experiments in (Huang et al., 2020) have demonstrated its low-rank properties.

$$\mathcal{P}(\mathbf{X}_j) \begin{bmatrix} \hat{\mathbf{p}}_j^r \\ \hat{\mathbf{p}}_j^i \end{bmatrix} = \left\{ \begin{bmatrix} \mathbf{X}_j^{r+} - \mathbf{X}_j^{r-} & \mathbf{X}_j^{i-} - \mathbf{X}_j^{i+} \\ \mathbf{X}_j^{i+} + \mathbf{X}_j^{i-} & -\mathbf{X}_j^{r-} - \mathbf{X}_j^{r+} \end{bmatrix} \right\} \begin{bmatrix} \hat{\mathbf{p}}_j^r \\ \hat{\mathbf{p}}_j^i \end{bmatrix} \approx \mathbf{0}. \quad \text{Eq. (D8)}$$

D: Reasons for choosing the Wilcoxon test

We use the overall quality scores of DONATE ($\mathbf{X}_1, \dots, \mathbf{X}_n, \dots, \mathbf{X}_{60}$) and PAIR ($\mathbf{Y}_1, \dots, \mathbf{Y}_n, \dots, \mathbf{Y}_{60}$) as an example to explain the appropriateness of choosing the Wilcoxon signed rank test, and the significance of results. Here is why the signed rank test is used:

- 1) The scores are a small sample data (60 data points) with unknown distribution, which is suitable to use a non-parametric test.
- 2) Each data point in the ordered sample is a pair of measurements: $(\mathbf{X}_1, \mathbf{Y}_1), \dots, (\mathbf{X}_n, \mathbf{Y}_n), \dots, (\mathbf{X}_{60}, \mathbf{Y}_{60})$, which can be converted to situation of one-sample test.

Then, our null and alternative hypotheses are:

H₀: The median of $\mathbf{X}_n - \mathbf{Y}_n$ is zero, which means there is no difference between DONATE and PAIR scores.

H₁: The median of $\mathbf{X}_n - \mathbf{Y}_n$ is not zero, which means there is difference between DONATE and PAIR scores.

After performing the Wilcoxon signed-test, we find a p -value < 0.001 , which means the difference between DONATE and PAIR is significant. Considering that the mean score of DONATE is also higher than that of PAIR, we can conclude that the four radiologists suggest that the overall quality of DWI images reconstructed by DONATE is significantly better than that of PAIR.

The other results of Wilcoxon signed-test in our reader study can be interpreted similarly.

References:

Haldar, J.P., 2013. Low-rank modeling of local k -space neighborhoods (LORAKS) for constrained MRI. *IEEE Transactions on Medical Imaging* 33, 668-681.

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Huang, Y., Zhang, X., Guo, H., Chen, H., Guo, D., Huang, F., Xu, Q., Qu, X., 2020. Phase-constrained reconstruction of high-resolution multi-shot diffusion weighted image. *Journal of Magnetic Resonance* 312, 106690.